

Mahmoud Alyosify

Specialised Computer Vision Engineer | Self-Supervised Representation Learning & Visual Retrieval | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

Computer Vision Engineer focused on **representation learning, self-supervised pre-training and visual retrieval**. Built a full **SimCLR** pipeline with a **42-experiment augmentation sweep** and a semi-supervised **SupCon** extension reaching **84.30% top-1 accuracy**, then shipped it as a deployable **ONNX + FAISS** retrieval service. Earlier work covers **video-based physiological signal extraction (rPPG)** — a contactless vital-sign system graded **A+** and taken to **Microsoft Imagine Cup 2023 semi-finals** — and **CNN classification over time–frequency spectrograms** for deepfake audio detection. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**. Strong on the part most vision demos skip: rigorous augmentation ablation, honest evaluation and deployment.

EDUCATION

- MSc in Artificial Intelligence** Sep 2025 – Nov 2026
Expected Nov 2026
Queen's University, Kingston, Ontario, Canada
- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - Coursework: **Deep Learning, Generative AI, Large Language Models, Cloud Computing, End-to-End ML Pipelines.**
- B.Sc. Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
GPA: 3.53 / 4.00
Assiut University, Egypt
- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation from video via **rPPG** and wavelet filtering. **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

Computer Vision	CNNs, Image Classification, Self-Supervised Learning, Contrastive Learning (SimCLR, SupCon), Representation & Embedding Learning, Image Retrieval, Data Augmentation Design, Transfer Learning, Video-Based Signal Extraction (rPPG), Spectrogram-Based Classification
Frameworks & Tools	PyTorch, TensorFlow, OpenCV, Scikit-learn, NumPy, Hugging Face, Librosa, Optuna, SHAP
Deployment	ONNX, FAISS Vector Search, Docker, FastAPI, Streamlit, AWS, Latency Profiling, Git/GitHub, Linux
Evaluation	Top-1 / Top-k Accuracy, Retrieval Metrics, Linear Probing, Ablation Studies, Augmentation Sweeps, Reproducible Experiment Design
Also	Large Language Models, Fine-Tuning (LoRA, QLoRA), Multimodal & Agentic Systems, Reinforcement Learning (PPO, DQN)
Programming	Python, C++, C, SQL, C#, MATLAB, Bash

SELECTED COMPUTER VISION PROJECTS

- SimCLR-Vision-SSL — Self-Supervised Contrastive Representation Learning** 2026
PyTorch · SimCLR · SupCon · ONNX · FAISS
- Implemented a complete **SimCLR** pipeline — augmentation stack, projection head, NT-Xent contrastive objective and downstream evaluation — trained without labels.
 - Ran a **42-experiment augmentation sweep** to isolate which transformations and combinations actually drive representation quality, rather than accepting published defaults.
 - Extended the pipeline with a semi-supervised **SupCon** objective, reaching **84.30% top-1 accuracy** and evaluating learned embeddings on both classification and **image retrieval**.
 - Exported the encoder to **ONNX** and served embeddings through a **FAISS** index for deployment-ready nearest-neighbour visual search.
- Vitalism Solution — Contactless Vital-Sign Estimation from Video** 2023
Computer Vision · rPPG · Wavelet Filtering · Signal Processing
- Built an AI system estimating vital signs in real time from ordinary video by extracting **remote photoplethysmography (rPPG)** signals from facial regions and denoising them with wavelet filtering — no contact sensors.
 - Graded **A+** as the undergraduate capstone and selected as a **Semi-Finalist at Microsoft Imagine Cup 2023.**
- AudioShield — CNN Deepfake Detection over Mel-Spectrograms** 2023
TensorFlow · CNNs · Librosa · Streamlit
- Trained a **CNN** to classify authentic versus synthetic audio from **mel-spectrogram** images, applying vision-style convolutional modelling to time–frequency representations.
 - Deployed as a Streamlit application with real-time prediction and automated PDF reporting, targeting fraud detection, digital forensics and media authentication.
- Automated Multimodal Agent — Document Understanding to Narrated Presentation** 2026

Multimodal AI · Document Parsing · Generative AI

- Built an autonomous pipeline that parses visual and textual document structure and regenerates it as a fully narrated presentation, coordinating extraction, layout generation and audio synthesis.

Additional Projects

- **HORUS Sentinel** — multi-agent intelligence platform correlating multi-source findings into a knowledge graph with a self-hosted fine-tuned LLM.
- **SCRAP** — supervised risk-prediction pipeline forecasting satellite collision risk **48+ hours** ahead.
- **Also:** Verbosity-Aware Decoding (MSc research) · Horus-OSINT (Llama-3-8B fine-tuning on AWS) · Diabetes Onset Prediction.

PROFESSIONAL EXPERIENCE

Machine Learning Instructor Jul 2025 – Sep 2025
National Telecommunication Institute (NTI) Cairo, Egypt
• Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks, with practical lab sessions throughout.

AI in Cybersecurity Intern Jun 2025 – Jul 2025
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt
• Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo, Egypt
• Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

Online Instructor — AI, Algorithms & Data Science 2020 – Present
Udemy & YouTube Remote
• Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

Machine Learning Specialization — DeepLearning.AI	AWS Academy — ML Foundations, ML for NLP, Data Engineering
HCIP – AI (Huawei)	ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)
Semi-Finalist — Microsoft Imagine Cup 2023	1st Place — Smart Cities Hackathon 2022
Digilians Presidential Scholarship — Queen's University	Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.